

Youssef Sawiris

Math Educator & Software Engineer

Oakland, CA · 510.331.7772 · sawirisyoussef@gmail.com

github.com/ysawiris · linkedin.com/in/youssef-sawiris · youtube.com/@learn-with-yoyo

SKILLS

Languages

Python · TypeScript · JavaScript ·
Scala · C++ · Go · Lua · Ruby ·
SQL · Bash

AI / Agents

Anthropic SDK · Claude Agent
SDK · MCP servers · OpenAI ·
Gemini · ElevenLabs · LLM tool-
use

Web

Next.js · React · Tailwind · Prisma ·
Node.js · Django · Rails ·
shadcn/ui

Teaching

Project-based learning · IEP-
aligned instruction · Small-group
facilitation · AI literacy

EDUCATION

Reach University

M.A. Education + Teaching
Credential (in progress)
Expected 2027 · Oakland, CA

Dominican University of California

B.S. Computer Science · Back-
end concentration
Jul 2021 · San Rafael, CA

SELECTED PROJECTS

Kordon

Solo Builder

Next.js e-commerce site for an
aquarium/pond pet-products
brand.

FlowAgent

Solo Builder

AI workflow automation across
Instagram, WhatsApp, Gmail, and
Calendar.

Satellite Tracker

Flask + Python satellite tracker.
Back-End Lead, team of 4.

PROFILE

Software engineer with 3+ years at Strava and a B.S. in Computer Science, now teaching 6th grade math in Oakland. My pedagogy leans on structured whole-class formative checks every lesson and IEP-aligned differentiation carried over from my prior special-education role at the same school. I write production Python and ship AI-agent products on the side, and I love turning dense technical ideas into hands-on projects—especially for students who don't see themselves in tech yet.

EXPERIENCE

Lodestar Public Charter School · 6th Grade Math Teacher

Jan 2025 - Present · Oakland, CA

- Teach 6th grade math to ~25 students per period, building computational thinking and problem-solving aligned to CA Common Core standards.
- Use project-based, small-group facilitation with constant checks for understanding; differentiate and write IEP-aligned adaptations from my prior SPED role at the same school.
- Work primarily with first-generation, low-income students of color in urban Oakland—the same scholar community SMASH serves.
- Run structured whole-class formative checks every lesson—mini-whiteboards, half-sheet quick-checks, and IM cool-downs—to surface confusion in real time and adjust pacing.

2CM · Software Engineering Consultant

Feb 2024 - Present · Oakland, CA

- Engaged on full-stack feature work for a multi-tenant Next.js / TypeScript SaaS for countertop fabricators.
- Recent engagements: canvas drawing tools, JSON pricing-rule schemas, Stripe Connect billing, and Three.js / React-PDF rendering for quotes and 3D visualizations.

Strava · Software Engineer I & II

Jun 2021 - Jan 2024 · San Francisco, CA

- Re-indexed the Strava Search ("Surtch") service to support 9B+ activities, improving query latency and unblocking new product features.
- Shipped the Best-Ever Efforts feature end-to-end, owning complex multi-system data backfills.
- Migrated the Garmin integration to a ping-and-pull pipeline processing 2M+ daily events.
- Led the Server Guild and mentored junior engineers on debugging, code review, and incident response.

NVIDIA · Product Engineering Technician

Dec 2015 - Aug 2016 · Santa Clara, CA

- Identified GPU production quality issues on optical and electro-mechanical lines; partnered with engineering on root-cause analysis and component-level corrective actions.
- Assembled, aligned, and bench-tested pre-production graphic processors against SLT and ESD industry standards.